



Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 3

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Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	11/7/26
Title of Assignment:	Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.
Problem Statement	Develop a Java application demonstrating method overloading and the use of static variables, methods in class design. Learning Objectives To implement method overloading and use static variables and methods in Java programs.
Theory	Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Program Code

```
1  import java.util.*;
2
3  class Calci {
4      static double add (double a, double b) {
5          return a + b;
6      }
7
8      static double add (double a, double b, double c) {
9          return a + b + c;
10     }
11 }
12
13 public class Main_1 {
14     public static void main (String[] args) {
15         try (Scanner in = new Scanner(System.in)) {
16             System.out.print("Enter first number : ");
17             int a = in.nextInt();
18
19             System.out.print("Enter second number : ");
20             int b = in.nextInt();
21
22             System.out.print("Enter third number : ");
23             int c = in.nextInt();
24
25             System.out.println("\nSum of two : " + Calci.add(a, b));
26             System.out.println("Sum of three : " + Calci.add(a, b, c));
27         }
28     }
29 }
30
31
```

2nd

```
1
2  class BankAccount {
3      void rate () {
4          System.out.println(x: "The roi is 4%");
5      }
6  }
7
8  class SavingsAccount extends BankAccount {
9      void rate () {
10         System.out.println(x: "The roi is 6%");
11     }
12 }
13
14 class CurrentAccount extends BankAccount {
15     void rate () {
16         System.out.println(x: "The roi is 8%");
17     }
18 }
19
20 public class Main_2 {
21     Run | Debug
22     public static void main (String[] args) {
23         CurrentAccount ca = new CurrentAccount();
24         ca.rate();
25     }
26 }
```

Output Screenshots

```

PS C:\Users\Wilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Wilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Main_1'
Enter first number : 10
Enter second number : 20
Enter third number : 50

Sum of two : 30.0
Sum of three : 80.0
PS C:\Users\Wilima\Documents\JAVA PROJECTS\LAB EXP 3>

PS C:\Users\Wilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Wilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Main_2'
The roi is 8%
PS C:\Users\Wilima\Documents\JAVA PROJECTS\LAB EXP 3>

```

Practice Questions

Q1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

```

1  import java.util.Scanner;
2
3  class Calculator {
4      static int count = 0;
5
6      int add (int a , int b) {
7          count++;
8          return a + b;
9      }
10
11     double add (double a, double b) {
12         count++;
13         return a + b;
14     }
15 }
16
17 public class Ex1 {
18     Run | Debug
19     public static void main (String[] args) {
20         try (Scanner in = new Scanner(System.in)) {
21             System.out.print(s: "Enter first integer : ");
22             int a = in.nextInt();
23
24             System.out.print(s: "Enter second integer : ");
25             int b = in.nextInt();
26
27             System.out.print(s: "Enter first decimal number : ");
28             double _a = in.nextDouble();
29
30             System.out.print(s: "Enter second decimal number : ");
31             double _b = in.nextDouble();
32
33             Calculator addition = new Calculator();
34             System.out.println("\nMethod One : Sum = " + addition.add(a, b));
35             System.out.println("Method two : Sum = " + addition.add(_a, _b));
36             System.out.println("Number of operations : " + Calculator.count);
37         }
38     }
39 }

```

Q2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```

J Ex2.java > Ex2
1  import java.util.Scanner;
2
3  class RestaurantBilling {
4      static int count = 0;
5
6      // Takeaway
7      int bill (double amount) {
8          count++;
9          return (int)(amount + 10);
10     }
11
12     // Dine-in
13     int bill (double amount, double service) {
14         count++;
15         return (int)(amount + service / 100 * amount);
16     }
17
18     // Delivery
19     int bill (double amount, double tax, double platform) {
20         count++;
21         return (int)(amount + platform / 100 * amount + tax / 100 * amount);
22     }
23 }

```

Q3. Method Overriding: A vehicle tracking system needs to calculate the maximum speed of different types of vehicles.

A generic vehicle has a base speed.

A car has a higher max speed.

A bike has an even higher max speed.

```

1  class Vehicle {
2      int speed() {
3          return 100;
4      }
5  }
6
7  class Car extends Vehicle {
8      int speed() {
9          return 200;
10     }
11 }
12
13 class Bike extends Vehicle {
14     int speed() {
15         return 300;
16     }
17 }
18
19 public class Ex3 {
20     public static void main (String[] args) {
21         Vehicle v = new Vehicle();
22         Car c = new Car();
23         Bike b = new Bike();
24
25         System.out.println("The speed for the vehicle is : " + v.speed() + " km\\hr");
26         System.out.println("The speed for the vehicle is : " + c.speed() + " km\\hr");
27         System.out.println("The speed for the vehicle is : " + b.speed() + " km\\hr");
28     }
29 }

```

Q4. Method Overloading: Payment Processing System

A company is building a payment gateway that allows users to pay using different methods:

Credit Card (card number & CVV)

Bank Transfer (account number & IFSC code)

UPI ID

Ex4.java > Ex4

```
1  import java.util.Scanner;
2
3  class Payment {
4      void pay(String card, String cvv) {
5          System.out.println(x: "Paid using credit card.");
6      }
7
8      void pay(long number, String ifsc) {
9          System.out.println(x: "Paid using Bank Transfer.");
10     }
11
12     void pay(String upi) {
13         System.out.println(x: "Paid using UPI ID.");
14     }
15 }
16
17 public class Ex4 {
18     public static void main (String[] args) {
19         try (Scanner in = new Scanner(System.in)) {
20             Payment paym = new Payment();
21
22             while (true) {
23                 System.out.println(x: "----- Payment -----");
24
25                 System.out.println(x: "Select Mode : ");
26                 System.out.println(x: "(1) Credit Card");
27                 System.out.println(x: "(2) Bank Transfer");
28                 System.out.println(x: "(3) UPI ID");
29                 System.out.println(x: "(4) Exit");
30                 System.out.print(s: "Enter choice : ");
31                 int choice = in.nextInt();
32
33                 if (choice == 4) {
34                     System.out.println(x: "\nExiting...");
35                     break;
36                 }
37
38                 if (choice == 1)
39                     paym.pay(card: "card number", cvv: "cvv");
40                 else if (choice == 2)
41                     paym.pay(number: 1234565432, ifsc: "cvv");
42                 else if (choice == 3)
43                     paym.pay(upi: "upi id");
44                 else
45                     System.out.println(x: "Wrong Choice!!!");
46
47                 System.out.println();
48             }
49         }
50     }
51 }
```

Output Screenshots

Q1:

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex1'
Enter first integer : 2
Enter second integer : 7
Enter first decimal number : 4.677
Enter second decimal number : 213.4134

Method One : Sum = 9
Method two : Sum = 218.0904
Number of operations : 2
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> |
```

Q2:

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex2'
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 1
Enter the amount : Rs.2000
The total amount is = Rs.2010

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 2
Enter the amount : Rs.2000
The total amount is = Rs.2200

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 1
Enter the amount : Rs.2000
The total amount is = Rs.2150

----- Billing -----
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 4

Exiting...

----- Total Orders -----
Total orders : 3
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> |
```

Q3:

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Program Files\Java\jdk-8.0.60\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex3'
The speed for the vehicle is : 100 km\hr
The speed for the vehicle is : 200 km\hr
The speed for the vehicle is : 300 km\hr
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> |
```

Q4:

```

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> & 'c:\Pro
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Nilima\AppData
edhat.java\jdt_ws\LAB EXP 3_8277c32b\bin' 'Ex4'
----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 1
Paid using credit card.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 2
Paid using Bank Transfer.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 3
Paid using UPI ID.

----- Payment -----
Select Mode :
(1) Credit Card
(2) Bank Transfer
(3) UPI ID
(4) Exit
Enter choice : 4

Exiting...
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 3> █

```

Conclusion

This assignment helped in understanding the concepts of method overloading and static variables and methods in Java. It demonstrated how method overloading enables multiple methods with the same name to perform different tasks based on their parameter lists, improving code flexibility and readability. The use of static variables and methods showed how data and functionality can be shared among all objects of a class, reducing memory usage and simplifying program design. These concepts are fundamental to object-oriented programming and are widely used in developing efficient, reusable, and well-structured Java application.